

SCADA / ICS Migration Continuity

2026, 2027 and beyond

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Static PDF companion

Use the matching XLSX for editable inventory records, formulas, drop-downs, evidence, owners and status. This PDF is a readable reference and is not an interactive workbook.

How to use this companion

Start with scope and accountable owners. Record the cryptographic dependency, confidentiality or authenticity lifetime, algorithm and implementation evidence. Prioritize exposed, long-lived, high-impact dependencies; confirm supported protocol and product paths; test interoperability, failure modes, rollback and operational impact before deployment.

Final standards, drafts, research selections and local planning targets are different statuses. NIST alignment means the material draws from the listed sources; it does not claim NIST endorsement or completeness of every future recommendation.

Workbook: [secureazcloud-scada-ics-pqc-continuity-checklist-v2.0.xlsx](#)

Included worksheets

[Start Here](#) | [Continuity Checklist](#) | [OT Crypto Inventory](#) | [Latency Test Cases](#) | [Vendor Roadmap](#) | [Sources](#)

Start Here

Read this section alongside the corresponding worksheet in the editable workbook. Record decisions and evidence there.

Public safety-first checklist for OT/ICS/SCADA cryptographic discovery, latency validation, legacy gateways, vendor dependencies, hybrid mode, rollback, maintenance windows, and safety continuity.

Version: v2.0

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Purpose: Use this checklist to convert SCADA/ICS PQC migration planning into operationally safe discovery, lab validation, maintenance-window, rollback, and continuity activities.

Safety-first use: No cryptographic migration in production OT should proceed without safety impact review, operations approval, validated rollback, vendor guidance, and tested compensating controls.

Evidence standard: Evidence should include OT asset inventory, protocol map, vendor roadmap, maintenance-window approval, lab test results, latency/performance baselines, rollback record, and safety review.

Redacted sharing: Do not publish real plant names, IP addresses, engineering diagrams, vendor support cases, firmware details, or safety-system configurations.

Caution: This is an implementation planning aid, not a substitute for engineering safety analysis, regulatory obligations, or vendor-supported product changes.

2026 baseline: Assign owners, discover public-key dependencies, assess data and signature lifetimes, confirm supplier versions and establish tested recovery.

2027 delivery: Pilot supported profiles, verify relying parties and operational limits, approve migration waves and require measurable supplier commitments.

2028 and later: Expand deployment based on risk and support. Track residual quantum-vulnerable paths, exceptions, replacements and standards updates.

Timeline interpretation: These horizons are suggested local planning phases. IR 8547 is draft. Algorithm-specific 2030/2035 proposals are not a universal mandate or a prediction of a quantum computer date.

Evidence workflow: Complete needs owner, evidence, review date and reviewer. Accepted Risk / Not Applicable also needs rationale and approval. Next action flags missing prerequisites.

Input legend: Amber cells are editable. P1/P2/P3 are local priorities, not NIST scores. Example inventory rows are fictional. Blank cells mean unassessed. No implementation results are asserted.

Update cadence: Recheck sources, vendor support, inventory evidence and exceptions quarterly and after material changes. This is a local cadence. Review new evidence before claiming alignment with future 2027+ guidance.

Controls: 37

Interpretation: Counts measure documentary prerequisites. They do not certify NIST compliance or independently validate implementation.

Continuity Checklist

Use each item with evidence, an accountable owner, a target date, and the editable workbook status fields. An unchecked control is not satisfied by this reference document.

01 Create joint IT/OT PQC migration governance with plant operations, safety, engineering, cybersecurity, and vendor support.

Domain: Governance

Safety or Operational Rationale: OT migration decisions affect safety, uptime, quality, and physical process continuity.

Priority: P1

Control ID: OT-C001

NIST sources: N05, N13

Planning horizon: 2026 baseline

Threat / gap: See task rationale

02 Define production change moratorium criteria, emergency rollback authority, and safety-stop conditions.

Domain: Governance

Safety or Operational Rationale: PQC pilots must not override safety operations or production continuity.

Priority: P1

Control ID: OT-C002

NIST sources: N05, N13

Planning horizon: 2026 baseline

Threat / gap: See task rationale

03 Inventory OT assets using public-key cryptography: remote access, engineering workstations, historians, gateways, HMIs, PLC/RTU management, vendor portals, and update mechanisms.

Domain: Discovery

Safety or Operational Rationale: Unknown cryptographic dependencies can become hidden blockers.

Priority: P1

Control ID: OT-C003

NIST sources: N05, N13

Planning horizon: 2026 baseline

Threat / gap: See task rationale

04 Map protocols and gateways: TLS, SSH, VPN, OPC UA, MQTT, DNP3 Secure Authentication, IEC 61850, Modbus/TCP wrappers, vendor tunnels, and file transfer mechanisms.

Domain: Discovery

Safety or Operational Rationale: Protocol constraints drive latency, certificate size, and compatibility testing.

Priority: P1

Control ID: OT-C004

NIST sources: N05, N13

Planning horizon: 2026 baseline

Threat / gap: See task rationale

05 Capture baseline latency, jitter, packet loss, timeout, CPU, memory, and bandwidth for critical flows before any PQC/hybrid test.

Domain: Latency

Safety or Operational Rationale: PQC and hybrid changes can affect message size and computational overhead.

Priority: P1

Control ID: OT-C005

NIST sources: N05, N13

Planning horizon: 2026 baseline

Threat / gap: See task rationale

06 Define maximum acceptable latency and failure thresholds for each control path, monitoring path, and maintenance path.

Domain: Latency

Safety or Operational Rationale: Test criteria must be engineering-approved, not security-only.

Priority: P1

Control ID: OT-C006

NIST sources: N05, N13

Planning horizon: 2026 baseline

Threat / gap: See task rationale

07 Identify gateways, terminal servers, protocol converters, serial-to-IP devices, VPN concentrators, and firewall appliances with embedded crypto.

Domain: Legacy Gateway

Safety or Operational Rationale: Embedded/legacy devices may not support algorithm changes without firmware or replacement.

Priority: P1

Control ID: OT-C007

NIST sources: N05, N13

Planning horizon: 2026 baseline

Threat / gap: See task rationale

08 Request vendor PQC roadmap, supported firmware versions, cryptographic disclosure, maintenance-window requirements, and rollback guidance.

Domain: Vendor Dependency

Safety or Operational Rationale: OT supplier support windows and certifications constrain safe migration sequencing.

Priority: P1

Control ID: OT-C008

NIST sources: N05, N13

Planning horizon: 2026 baseline

Threat / gap: See task rationale

09 Determine where hybrid classical/PQC mode can be tested without affecting deterministic control loops.

Domain: Hybrid Mode

Safety or Operational Rationale: Hybrid mode may be useful during transition but must be tested for interoperability and performance.

Priority: P2

Control ID: OT-C009

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: See task rationale

10 Map change windows to production cycles, safety permits, staffing, vendor availability, and rollback duration.

Domain: Maintenance Windows

Safety or Operational Rationale: OT changes often require constrained windows and onsite expertise.

Priority: P1

Control ID: OT-C010

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: See task rationale

11 Create and rehearse rollback for certificates, firmware, gateway configuration, firewall rules, remote access policy, trust stores, and monitoring baselines.

Domain: Rollback

Safety or Operational Rationale: Rollback must restore safe and stable operation, not just application connectivity.

Priority: P1

Control ID: OT-C011

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: See task rationale

12 Build an OT-representative lab test using equivalent firmware, gateways, certificates, protocol paths, and monitoring tools.

Domain: Lab Validation

Safety or Operational Rationale: Production OT should not be the first place PQC-related behavior is observed.

Priority: P1

Control ID: OT-C012

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: See task rationale

13 Maintain compensating controls where PQC cannot be deployed: segmentation, allowlisting, jump hosts, strong monitoring, encryption overlays, maintenance access controls, and vendor contractual controls.

Domain: Compensating Controls

Safety or Operational Rationale: Legacy assets may remain in service for extended support windows.

Priority: P2

Control ID: OT-C013

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: See task rationale

14 Update monitoring for handshake failures, certificate validation errors, VPN failures, remote access anomalies, and gateway performance changes.

Domain: Monitoring

Safety or Operational Rationale: Operational visibility is essential during pilot and rollback.

Priority: P2

Control ID: OT-C014

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: See task rationale

15 Add PQC migration failure scenarios to OT incident and outage response runbooks.

Domain: Incident Response

Safety or Operational Rationale: Crypto migration can create authentication, availability, and remote access incidents.

Priority: P2

Control ID: OT-C015

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: See task rationale

16 Create lifecycle plan for unsupported devices, firmware end-of-life, vendor roadmap gaps, and compensating controls through decommissioning.

Domain: Lifecycle

Safety or Operational Rationale: Some OT assets cannot migrate quickly and need managed exceptions.

Priority: P1

Control ID: OT-C016

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: See task rationale

17 Use passive observation and vendor-approved offline/maintenance-window discovery. Obtain operations approval before active probing.

Domain: Safe discovery

Safety or Operational Rationale: Discovery must not disturb deterministic or fragile systems.

Priority: P1

Control ID: OT-C017

NIST sources: N07, N13

Planning horizon: 2026 baseline

Threat / gap: Disruption from discovery

18 Assign stable asset/flow IDs. Record hardware, firmware, library/provider, algorithm/parameter, key storage and observation date.

Domain: Inventory quality

Safety or Operational Rationale: Support and replacement decisions need version-level evidence.

Priority: P1

Control ID: OT-C018

NIST sources: N05, N07, N13

Planning horizon: 2026 baseline

Threat / gap: Hidden or stale dependencies

19 Assess confidentiality lifetime and capture exposure for recipes, engineering IP, remote-access traffic and archives.

Domain: Threat assessment

Safety or Operational Rationale: Harvest-now-decrypt-later exposure can outlast long asset lifecycles.

Priority: P1

Control ID: OT-C019

NIST sources: N06, N07

Planning horizon: 2026 baseline

Threat / gap: Harvest now, decrypt later

20 Inventory secure boot, firmware signatures, update packages, engineering project signing and long-term verification.

Domain: Authenticity

Safety or Operational Rationale: Future forgery can affect device integrity and maintenance trust.

Priority: P1

Control ID: OT-C020

NIST sources: N02, N03, N13

Planning horizon: 2026 baseline

Threat / gap: Future signature forgery

21 Separate encrypted, authenticated-only and unauthenticated legacy flows. Record actual cryptography and gateway boundaries.

Domain: Protocol mapping

Safety or Operational Rationale: Protocol names alone do not establish encryption or PQC protection.

Priority: P1

Control ID: OT-C021

NIST sources: N07, N13

Planning horizon: 2026 baseline

Threat / gap: False protection assumptions

22 Map ML-KEM to key establishment and ML-DSA/SLH-DSA to signatures. Select supported profiles and measured parameters.

Domain: Algorithm selection

Safety or Operational Rationale: Final algorithms do not guarantee OT protocol or product support.

Priority: P1

Control ID: OT-C022

NIST sources: N01, N02, N03, N04

Planning horizon: 2027 pilots and rollout

Threat / gap: Unsupported primitives or profiles

23 Map safety systems, fail-safe states, interlocks and deterministic loops. Record independent safety review and prohibited changes.

Domain: Safety boundaries

Safety or Operational Rationale: Crypto changes must preserve the approved safety function.

Priority: P1

Control ID: OT-C023

NIST sources: N13

Planning horizon: 2026 baseline

Threat / gap: Unsafe process or safety-function loss

24 Measure handshake rate, certificate size, CPU/RAM, fragmentation, packet loss and failover under worst credible load.

Domain: Resource limits

Safety or Operational Rationale: Average latency can hide timing and availability failures.

Priority: P1

Control ID: OT-C024

NIST sources: N08, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: Resource exhaustion or timing failure

25 Test power loss, cold start, disconnected operation, time-sync loss and offline validation in representative configurations.

Domain: Recovery

Safety or Operational Rationale: OT can remain isolated or offline for long periods.

Priority: P1

Control ID: OT-C025

NIST sources: N09, N13, N15

Planning horizon: 2027 pilots and rollout

Threat / gap: Recovery or offline authentication failure

26 Test expiry, renewal, revocation, trust-anchor distribution and key recovery across redundant/intermittent sites.

Domain: Key lifecycle

Safety or Operational Rationale: Trust failures can disable maintenance or process visibility.

Priority: P1

Control ID: OT-C026

NIST sources: N05, N09, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: Expired credentials or lost keys

27 Verify new and historical signatures, reject tampering and unauthorized downgrade, and document approved recovery media.

Domain: Firmware authenticity

Safety or Operational Rationale: Rollback must not authorize unauthenticated or unsafe software.

Priority: P1

Control ID: OT-C027

NIST sources: N02, N03, N13, N15

Planning horizon: 2027 pilots and rollout

Threat / gap: Forged firmware or unsafe downgrade

28 Record vendor-supported construction, negotiation and downgrade policy. Test component failures and capture algorithms.

Domain: Hybrid policy

Safety or Operational Rationale: Hybrid adoption is protocol-dependent and can retain vulnerable paths.

Priority: P1

Control ID: OT-C028

NIST sources: N04, N05

Planning horizon: 2027 pilots and rollout

Threat / gap: Downgrade or insecure composition

29 Document cleartext segments, trust termination and residual risk behind gateways. Test overlay failure and failover.

Domain: Gateway boundary

Safety or Operational Rationale: An overlay protects only its defined path and does not repair endpoint trust.

Priority: P1

Control ID: OT-C029

NIST sources: N13

Planning horizon: 2027 pilots and rollout

Threat / gap: Exposed legacy segment or single point of failure

30 Obtain exact supported versions, release commitments, test evidence, patch policy, end of support and safety/certification impact.

Domain: Supplier evidence

Safety or Operational Rationale: Roadmaps must be tied to a deployable, supportable product.

Priority: P1

Control ID: OT-C030

NIST sources: N29

Planning horizon: 2026 baseline

Threat / gap: Supplier delay or unsupported lifecycle

31 Where required, verify CMVP certificate, module/version, operating environment, approved mode, PQC service coverage and current status.

Domain: Module assurance

Safety or Operational Rationale: Algorithm approval is not module/product validation. Historical status is not revocation.

Priority: P1

Control ID: OT-C031

NIST sources: N24, N25, N26

Planning horizon: 2027 pilots and rollout

Threat / gap: Unsupported assurance claim

32 Require approved engineering limits, representative tests, safety/operations approval and vendor-supported rollback before release.

Domain: Lab exit review

Safety or Operational Rationale: Passing connectivity tests alone cannot authorize an OT change.

Priority: P1

Control ID: OT-C032

NIST sources: N05, N13

Planning horizon: 2027 pilots and rollout

Threat / gap: Premature production change

33 Include device configs, firmware, logic, trust stores, certificates and recovery credentials in protected backups. Test restoration and document dependencies.

Domain: Backup governance

Safety or Operational Rationale: Backup availability is only useful when the safe process state can be restored.

Priority: P1

Control ID: OT-C033

NIST sources: N13, N15

Planning horizon: 2026 baseline

Threat / gap: Unrecoverable process state

34 Record owner, risk approver, controls, review/expiry date, replacement milestone and decommission evidence for unsupported assets.

Domain: Exception management

Safety or Operational Rationale: Residual risk needs review through long asset lifecycles.

Priority: P1

Control ID: OT-C034

NIST sources: N13, N27

Planning horizon: 2028+ continuous migration

Threat / gap: Permanent unmanaged exposure

35 Reconcile after firmware updates, plant expansion, cloud/IIoT integration or vendor access changes. Retest critical flows.

Domain: Continuous assurance

Safety or Operational Rationale: Later operational changes can invalidate previous migration results.

Priority: P1

Control ID: OT-C035

NIST sources: N05, N13; N14 draft watch

Planning horizon: 2028+ continuous migration

Threat / gap: Drift or silent fallback

36 Monitor draft SP 800-82 Rev. 4 (2026-09-21), IR 8547, key-management/transition drafts, SP 800-230, SP 800-133 Rev. 3 and pending HQC/FN-DSA.

Domain: Standards watch

Safety or Operational Rationale: SP 800-82 Rev. 3 remains the final OT baseline. Draft proposals and future dates are not mandates.

Priority: P1

Control ID: OT-C036

NIST sources: N06, N10, N12, N14, N19-N21, N30, N31

Planning horizon: 2028+ continuous migration

Threat / gap: Stale standards or premature adoption

37 Confirm migration or safe retirement of legacy keys, trust anchors, gateways, firmware verification and recovery-media dependencies.

Domain: Retirement

Safety or Operational Rationale: Residual dependencies preserve exposure after the primary path migrates.

Priority: P1

Control ID: OT-C037

NIST sources: N05, N09, N13

Planning horizon: 2028+ continuous migration

Threat / gap: Residual quantum-vulnerable paths

OT Crypto Inventory

The editable workbook contains the full data-entry template. The field structure below identifies information to collect. Example records are illustrative and are not a validated production inventory.

Fields in the editable worksheet

01. Site/Zone
02. Asset / System
03. Function
04. Protocol / Path
05. Crypto Use
06. Algorithm / Key Type
07. Legacy Gateway?
08. Safety Criticality
09. Latency Tolerance
10. Maintenance Window
11. Vendor Roadmap
12. Hybrid Mode Plan
13. Rollback Method
14. Compensating Control
15. Priority
16. Evidence Link
17. Notes
18. Asset / flow ID
19. Hardware and firmware build
20. Crypto library/provider version
21. Key establishment / signature role
22. Confidentiality lifetime (years)

- 23. Signature verification lifetime
- 24. Discovery method / approval
- 25. Last observed
- 26. Owner
- 27. End of support
- 28. Replacement milestone
- 29. Exception approver / review date
- 30. Residual exposure after gateway

Illustrative records - selected fields

01 Remote access gateway

Function: Vendor maintenance access

Protocol / Path: VPN / TLS

Crypto Use: Key establishment and certificate auth

Algorithm / Key Type: ECDHE_RSA / RSA-2048

Notes: Example only

Asset / flow ID: EX-OT-001

02 Historian connector

Function: Data collection

Protocol / Path: TLS / MQTT

Crypto Use: TLS server and client auth

Algorithm / Key Type: ECDSA / P-256

Notes: Example only

Asset / flow ID: EX-OT-002

Latency Test Cases

Capture a baseline and explicit acceptance criteria, test in the approved environment, and preserve result and rollback evidence before authorizing a rollout.

01 TLS/mTLS gateway handshake under expected maintenance traffic

Test ID: OT-001

Metric: Handshake success, latency, CPU, memory

Baseline: Capture before change

Acceptance Criteria: No critical errors; latency within engineering threshold; CPU/memory within safety-approved limit.

Test Environment: Lab

02 Certificate chain-size and trust-store compatibility test

Test ID: OT-002

Metric: Chain validation success, error rate, log visibility

Baseline: Current chain size and validation behavior

Acceptance Criteria: All test clients validate; failures are observable and reversible.

Test Environment: Lab

03 Legacy gateway firmware/configuration rollback

Test ID: OT-003

Metric: Rollback time, configuration integrity, restored service

Baseline: Current gateway config backup

Acceptance Criteria: Rollback completes within approved window with service restored.

Test Environment: Lab

04 Vendor remote-access session through controlled jump path

Test ID: OT-004

Metric: Connection success, latency, logging, MFA/allowlist status

Baseline: Current vendor access baseline

Acceptance Criteria: No bypass of remote-access controls; logs retained.

Test Environment: Lab or maintenance window

05 Monitoring and alerting validation

Test ID: OT-005

Metric: Alerts for cert errors, auth failures, VPN failures, gateway CPU/memory

Baseline: Current monitoring baseline

Acceptance Criteria: Alerts fire for failure scenarios without unacceptable noise.

Test Environment: Lab

06 Hybrid mode interoperability trial where vendor-supported

Test ID: OT-006

Metric: Protocol success, latency, error rate, fallback behavior

Baseline: Current production-equivalent behavior

Acceptance Criteria: Hybrid mode does not break test flows; rollback is validated.

Test Environment: Lab

07 Worst credible load and degraded network

Test ID: OT-007

Metric: Tail latency, jitter, CPU, RAM and timeouts

Baseline: Production-equivalent baseline

Acceptance Criteria: All engineering-approved limits hold under peak load and packet loss.

Test Environment: Representative lab

08 Power loss, cold start and disconnected operations

Test ID: OT-008

Metric: Boot, local control, authentication and time sync

Baseline: Approved cold-start behavior

Acceptance Criteria: Required safe local operation and recovery remain available.

Test Environment: Representative lab

09 Tampered firmware and unauthorized downgrade

Test ID: OT-009

Metric: Signature validation and recovery-media trust

Baseline: Approved firmware/trust anchors

Acceptance Criteria: Invalid packages are rejected. Approved recovery restores a safe supported version.

Test Environment: Representative lab

10 Expiry, renewal and revocation while disconnected

Test ID: OT-010

Metric: Trust availability, logging and recovery

Baseline: Current lifecycle behavior

Acceptance Criteria: Expected policy holds without bypass; renewal/recovery works across disconnected sites.

Test Environment: Representative lab

11 Gateway failover and exposed-segment review

Test ID: OT-011

Metric: Failover time and protected path boundary

Baseline: Approved topology and baseline

Acceptance Criteria: No unapproved route or bypass. Failover meets engineering limits.

Test Environment: Representative lab

12 Malformed ciphertext and downgrade behavior

Test ID: OT-012

Metric: Failure, crash resistance and algorithms

Baseline: Vendor-approved behavior

Acceptance Criteria: Vetted implementation follows policy without secret exposure or silent fallback.

Test Environment: Representative lab

13 Safety and operator visibility during crypto failure

Test ID: OT-013

Metric: Interlocks, fail-safe state, HMI and alarms

Baseline: Approved safety/operations behavior

Acceptance Criteria: Independent engineering review confirms safety functions and required visibility.

Test Environment: Representative lab

14 Backup, trust-anchor and key recovery

Test ID: OT-014

Metric: Restore time, config integrity and key access

Baseline: Verified offline recovery set

Acceptance Criteria: Complete recovery meets approved safety/restoration limits with evidence.

Test Environment: Representative lab

15 Long-running soak and reconnect storm

Test ID: OT-015

Metric: Timing drift, memory, retries and continuity

Baseline: Production-equivalent steady state

Acceptance Criteria: No unacceptable timing, leak or loss of required control over the defined duration.

Test Environment: Representative lab

16 Post-upgrade regression and rollback

Test ID: OT-016

Metric: Critical flows, vendor access and inventory

Baseline: Last approved configuration

Acceptance Criteria: Critical paths and safe rollback are verified; inventory matches deployed version.

Test Environment: Representative lab

Vendor Roadmap

Read this section alongside the corresponding worksheet in the editable workbook. Record decisions and evidence there.

Fields in the editable worksheet

01. Vendor/Product
02. OT Function
03. Firmware/Version
04. Support Window
05. PQC Roadmap
06. Cryptographic Disclosure Available?
07. Maintenance Window Requirement
08. Rollback Guidance Available?
09. Owner
10. Evidence Link
11. Notes
12. PQC algorithms / parameters
13. Protocol profile / status
14. Release date commitment
15. CMVP certificate/version if required
16. Safety certification impact
17. Verified interoperability evidence
18. Support end date
19. Replacement plan / next review

Illustrative records - selected fields

01 Remote access gateway

Notes: Example only

PQC algorithms / parameters: Unknown

Protocol profile / status: Unknown

02 Legacy protocol gateway

Notes: Example only

PQC algorithms / parameters: Unknown

Protocol profile / status: Unknown

Sources

Official sources, publication status, and applicability recorded in the workbook. Recheck draft status, errata, and implementation guidance before a production decision.

01 FIPS 203 ML-KEM; key establishment, parameter sets 512/768/1024; monitor potential errata

Source ID: N01

Status as of 2026-09-29: Final 2024-08-13; notice 2025-11-17

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/fips/203/final>

02 FIPS 204 ML-DSA; signature generation and verification; monitor potential errata

Source ID: N02

Status as of 2026-09-29: Final 2024-08-13; notice 2026-07-31

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/fips/204/final>

03 FIPS 205 SLH-DSA; stateless hash-based signatures

Source ID: N03

Status as of 2026-09-29: Final 2024-08-13

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/fips/205/final>

04 SP 800-227; conforming KEM implementation, randomness, input validation, key destruction, authentication, key derivation/confirmation and multi-algorithm construction

Source ID: N04

Status as of 2026-09-29: Final 2025-09-18

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/227/final>

05 CSWP 39-upd1; governance, inventory, risk-prioritized agility, negotiation, supply chains, APIs, continuous improvement

Source ID: N05

Status as of 2026-09-29: Final 2025-12-19; updated 2026-06-29

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/cswp/39/upd1/considerations-for-achieving-crypto-agility/final>

06 IR 8547; HNDL, signature and code-verification distinctions, draft transition tables

Source ID: N06

Status as of 2026-09-29: Initial public draft 2024-11-12

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/ir/8547/ipd>

07 NCCoE Migration to PQC; discovery/inventory and interoperability workstreams; evolving output model

Source ID: N07

Status as of 2026-09-29: Active project; preliminary documents

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://www.nccoe.nist.gov/applied-cryptography/migration-to-pqc>

08 SP 1800-38 B/C; discovery and interoperability/performance demonstration

Source ID: N08

Status as of 2026-09-29: Initial preliminary draft 2023-12-19

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: [https://csrc.nist.gov/pubs/sp/1800/38/iprd-\(1\)](https://csrc.nist.gov/pubs/sp/1800/38/iprd-(1))

09 SP 800-57 Part 1 Rev. 5; key management baseline

Source ID: N09

Status as of 2026-09-29: Final 2020-05-04

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final>

10 SP 800-57 Part 1 Rev. 6; PQC-aware key management and storage proposals

Source ID: N10

Status as of 2026-09-29: Initial public draft 2025-12-05

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/57/pt1/r6/ipd>

11 SP 800-131A Rev. 2; current final algorithm transition baseline

Source ID: N11

Status as of 2026-09-29: Final 2019-03-21

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/131/a/r2/final>

12 SP 800-131A Rev. 3; proposed algorithm/key-strength transitions

Source ID: N12

Status as of 2026-09-29: Initial public draft 2024-10-21

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/131/a/r3/ipd>

13 SP 800-82 Rev. 3; OT performance, reliability, safety and controls

Source ID: N13

Status as of 2026-09-29: Final 2023-09-28

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

14 SP 800-82 Rev. 4; CSF2 alignment, asset management, monitoring, OT/IIoT/cloud, system-management security

Source ID: N14

Status as of 2026-09-29: Initial public draft 2026-09-21

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/82/r4/ipd>

15 SP 1339; backups in OT change and recovery practice

Source ID: N15

Status as of 2026-09-29: Final 2026-06-17

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/1339/final>

16 IR 8587; signed tokens/assertions, key lifecycle, verification, workload/automated identity token use, Section 7.7 PQC interoperability constraints

Source ID: N16

Status as of 2026-09-29: Final 2026-09-15

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/ir/8587/final>

17 SP 800-228-upd1; API lifecycle risks and controls

Source ID: N17

Status as of 2026-09-29: Final; original 2025-06-27, updated 2026-03-13

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/228/upd1/final>

18 CSWP 48; mapping PQC discovery and interoperability capabilities to risk frameworks

Source ID: N18

Status as of 2026-09-29: Initial public draft 2025-09-18

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/cswp/48/mapping-migration-to-pqc-project-capabilities-to-r/ipd>

19 NIST PQC project; current standardized/selected algorithm status

Source ID: N19

Status as of 2026-09-29: Project updated 2026-08-05

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/projects/post-quantum-cryptography>

20 HQC selection announcement; diversification backup, continue current migration, expected 2027 finalization

Source ID: N20

Status as of 2026-09-29: Selection announcement 2025-03-11, forecast not approval

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://www.nist.gov/news-events/news/2025/03/nist-selects-hqc-fifth-algorithm-post-quantum-encryption>

21 Additional digital signature schemes; Round3 announced May14,2026

Source ID: N21

Status as of 2026-09-29: Evaluation candidates, not standards

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/projects/pqc-dig-sig>

22 SP 800-204A; service-mesh security architecture

Source ID: N22

Status as of 2026-09-29: Final 2020-05-27

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/204/a/final>

23 SP 800-204B; service-mesh authentication/authorization

Source ID: N23

Status as of 2026-09-29: Final 2021-08

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/204/b/final>

24 CAVP program; algorithm validation is prerequisite to, not replacement for, module validation

Source ID: N24

Status as of 2026-09-29: Official program; updated 2026-09-22

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/projects/cryptographic-algorithm-validation-program>

25 CMVP FAQ; module/version/operating-environment verification, protocol/product validation boundary, FIPS140-2 active-list transition

Source ID: N25

Status as of 2026-09-29: Official FAQ checked 2026-09-29

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/Projects/cryptographic-module-validation-program/faqs>

26 CMVP validated modules; inspect certificate/security policy and active/historical/revoked status

Source ID: N26

Status as of 2026-09-29: Live certificate registry checked 2026-09-29

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/projects/cryptographic-module-validation-program/validated-modules>

27 CSWP 29 / CSF 2.0; cybersecurity outcome framework, not a prescriptive PQC implementation standard

Source ID: N27

Status as of 2026-09-29: Final 2024-02-26

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/cswp/29/the-nist-cybersecurity-framework-csf-20/final>

28 SP 800-53 Rev. 5; current control catalog is Release 5.2.0 referenced by publication-page planning note

Source ID: N28

Status as of 2026-09-29: Final; base 2020-09/update 2020-12-10; minor Release 5.2.0 finalized 2025-08-27

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final>

29 SP 800-161 Rev. 1-upd1; supplier risk and dependency governance

Source ID: N29

Status as of 2026-09-29: Final; original 2022-05, update 2024-11-01

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/161/r1/upd1/final>

30 SP 800-230; additional limited-signature SLH-DSA parameter sets, software/firmware/certificate watch item

Source ID: N30

Status as of 2026-09-29: Initial public draft 2026-04-13

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/230/ipd>

31 SP 800-133 Rev. 3; proposed PQC key generation, seed expansion, KEM and HSM considerations

Source ID: N31

Status as of 2026-09-29: Initial public draft 2026-04-17

Interpretation: Derived controls and annual phases are local planning recommendations. Drafts are not final requirements.

Public URL: <https://csrc.nist.gov/pubs/sp/800/133/r3/ipd>

32 CISA - Product Categories for Technologies That Use Post-Quantum Cryptography Standards

Source ID: C1

Status as of 2026-09-29: Supplementary guidance; see publication

Interpretation: 2026 product-category reference for PQC adoption, including cloud services, web software, and endpoint security categories.

Public URL: <https://content.govdelivery.com/accounts/USDHSCISA/bulletins/405c246>

33 CISA/NSA/NIST - Quantum-Readiness: Migration to Post-Quantum Cryptography

Source ID: C2

Status as of 2026-09-29: Supplementary guidance; see publication

Interpretation: Quantum-readiness roadmap, cryptographic inventory, vendor engagement, and supply-chain dependency assessment.

Public URL:

<https://www.cisa.gov/resources-tools/resources/quantum-readiness-migration-post-quantum-cryptography>

34 CISA - Post-Quantum Considerations for Operational Technology

Source ID: C3

Status as of 2026-09-29: Supplementary guidance; see publication

Interpretation: OT/ICS/SCADA migration considerations for safety, performance, vendor dependency, and operational continuity.

Public URL: <https://www.cisa.gov/resources-tools/resources/post-quantum-considerations-operational-technology>